

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Produced and validated **actuarial liability data** and **economic scenarios for IFRS and Solvency II reporting** using **Python, SQL**, and **Power BI** dashboards across enterprise systems; automated scenario generation reducing manual validation effort by **35%** and improving month-end close timeliness by **5 days**.
- Developed **actuarial reserves production pipelines** and **solvency capital calculations** using **Python ETL, PostgreSQL**, and **Jenkins CI/CD**; supported **IFRS movement analysis** and **solvency projections** for regulatory reporting, ensuring **100% accuracy** across quarterly ORSA and Planning Dialogue submissions.
- Streamlined and automated **actuarial reporting processes** via **Python scripting, REST API integrations**, and **data warehousing pipelines**; developed **Power BI dashboards** and **DAX measures** for actuarial communications to senior management and board committees, reducing report generation time by **40%**.
- Supported **new model developments** and **assumption setting exercises** through data validation, **statistical analysis** in **R**, and cross-functional collaboration with actuarial teams; applied **regression analysis** and **hypothesis testing** to underpin mortality, lapse, and expense assumptions for product repricing.
- Managed **change control** for actuarial system configuration updates, **UAT coordination**, and post-implementation support; facilitated actuarial **training materials** and onboarding documentation for new reporting frameworks, ensuring stakeholder alignment across cross-border European business operations.
- Assisted in **development and implementation of Solvency II risk function deliverables** including **Standard Model and Internal Model capital calculations, movement analysis, and risk management communications**; produced **risk dashboards** and **risk committee papers** using **Power BI** and **Tableau**, supporting ongoing maintenance of risk frameworks.

SKILLS

Actuarial & Financial Reporting – Actuarial liability data production and validation, IFRS reporting, Solvency II reporting, economic scenario generation, actuarial reserves production, IFRS movement analysis, solvency capital projections, ORSA Recovery Plan support, assumption setting, Internal Model support.

Risk Management & Capital Modelling – Solvency II Standard Model, Solvency II Internal Model, capital calculations, movement analysis, risk dashboards, risk committee reporting, planning dialogue support.

Data Engineering & Analytics – Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL, Oracle), ETL pipeline development, data warehousing, statistical analysis, regression analysis, hypothesis testing, data validation.

Business Intelligence & Dashboarding – Power BI (dashboards, DAX measures, data modelling, Power Query), Tableau, executive reporting, KPI tracking, data visualisation.

Software Engineering & DevOps – Java Spring Boot, REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker containerisation, Jenkins CI/CD pipelines, Git version control, Linux shell scripting.

Domain & Regulatory Knowledge – Life insurance product design and pricing, financial guarantees, hedging strategy, cross-border European operations, regulatory compliance, change management, UAT coordination.

Certifications & Education – MSc Data Analytics (NCI, Dublin); BE Computer Science (SNS College, India); IELTS Band 7/9; Full Stack Java Spring Boot & Angular; Web Development (UC Davis); Python Programming (University of Michigan).

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Actuarial Reserves & Solvency II Capital Automation Platform (HCL Technologies)

- Architected end-to-end **Python ETL pipeline** for automated **actuarial liability data production**, **economic scenario generation**, and **Solvency II capital calculations** integrating **PostgreSQL** data warehouse with **REST API** connections to legacy ERP systems; reduced month-end actuarial close cycle from **12 days to 7 days**.
- Developed **Power BI dashboard suite** with **DAX measures** for real-time **IFRS movement analysis**, **solvency projections**, and **assumption tracking**; published **interactive reports** enabling actuarial leadership to monitor reserve adequacy and capital position across **8 European jurisdictions**.
- Implemented **Jenkins CI/CD automation** for monthly scenario batch jobs, **data validation checks**, and **audit trail logging**; introduced **R statistical validation** routines for **hypothesis testing** on assumption changes, ensuring **100% accuracy** on IFRS and Solvency II submissions to regulators.
- Collaborated with cross-functional actuarial and risk teams to document **assumption setting exercises** and **model methodology**; produced **actuarial communications** (committee papers, reports, dashboards) explaining reserve movements and capital drivers to board and senior management.
- Supported **Internal Model development and maintenance** through data integration, scenario stress-testing, and **change control** procedures; coordinated **UAT** across actuarial and IT stakeholders, reducing post-implementation defects by **30%**.

Enterprise Payments Operations & SWIFT Messaging Analytics Dashboard (HCL Technologies)

- Built real-time **operational monitoring dashboard** using **Splunk** and **Power BI** tracking **SWIFT message flows** (MT/MX formats), payment transaction volumes, and **ISO 20022 compliance metrics**; enabled operations teams to visualize **SEPA and cross-border payment SLAs** with **99.8% uptime**.
- Automated **ETL pipelines** in **Python** for reconciliation of **payments processing workflows** across **PostgreSQL and Oracle** databases; validated **ISO 20022 message structure integrity** and flagged regulatory anomalies, supporting **GDPR and Data Protection compliance**.
- Produced **P1/P2 incident response reports** and **actuarial-style movement analysis** for payment operations; collaborated with banking-domain business analysts and compliance teams on **PSD2 and Open Banking** requirement mapping, reducing post-incident triage time by **25%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)